

Solution Sheet: Markov Chains & Dynamic Programming (DPP/DP)

How to use this sheet. For each exercise, I first restate the key idea(s), then carry out the computation step by step. All probabilities are exact unless a numerical approximation is explicitly stated.

Easy (9)

1. One-step and two-step transitions.

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.1, Fig. 5.1, book p. 106.

From Fig. 5.1 the transition matrix is

$$P = \begin{pmatrix} 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0.25 & 0.25 \\ 0 & 0.4 & 0.4 & 0.2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- One step:

$$\mathbb{P}(X_1 = 4 \mid X_0 = 1) = p_{14} = 0.5.$$

- Two steps:

$$\mathbb{P}(X_2 = 4 \mid X_0 = 1) = (P^2)_{14} = \sum_{j=1}^4 p_{1j}p_{j4}.$$

Only $j = 2$ and $j = 4$ matter (since $p_{11} = p_{13} = 0$), hence

$$(P^2)_{14} = p_{12}p_{24} + p_{14}p_{44} = 0.5 \cdot 0.25 + 0.5 \cdot 1 = 0.125 + 0.5 = \frac{5}{8} = 0.625.$$

2. State classification (quick).

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.1, “Classification of states”, book p. 106.

From Fig. 5.1 (or the matrix), state 4 has $p_{44} = 1$ and $p_{4j} = 0$ for $j \neq 4$, so:

- **Absorbing state(s):** state 4.
- **A communicating class:** $\{1, 2, 3\}$. Indeed, $1 \rightarrow 2$ and $2 \rightarrow 1$ in one step, and $2 \leftrightarrow 3$ as well (e.g. $2 \rightarrow 3$ with prob. 0.25 and $3 \rightarrow 2$ with prob. 0.4), hence all three communicate.

- **Transient vs recurrent:** $\{4\}$ is a closed class (absorbing), hence recurrent. States 1, 2, 3 are *transient* because they can reach 4 with positive probability, but 4 cannot reach them.

3. Gambler's ruin (small bankroll).

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.1, "Gambler's ruin problem", book p. 107.

Let M be the player who wins each round with probability $p = \frac{2}{3}$ (and loses with $q = \frac{1}{3}$). Total money is 3, so define the Markov chain $S_t \in \{0, 1, 2, 3\}$ = dollars held by M after t rounds. We start at $S_0 = 1$ and want $\mathbb{P}(\text{hit 3 before 0})$.

For the biased ruin problem ($p \neq q$), the absorption probability is

$$\mathbb{P}_i(\text{hit } N \text{ before } 0) = \frac{1 - (q/p)^i}{1 - (q/p)^N}.$$

Here $i = 1$, $N = 3$, and $q/p = (1/3)/(2/3) = 1/2$. Therefore,

$$\mathbb{P}(M \text{ ends with all \$3}) = \frac{1 - (1/2)^1}{1 - (1/2)^3} = \frac{1/2}{1 - 1/8} = \frac{1/2}{7/8} = \frac{4}{7}.$$

4. Waiting time for a pattern (HHH).

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.1, "Coin triplets, Part A", book pp. 109–110.

Use a Markov chain whose state is the length of the current run of consecutive heads.

Let

$$E_0 = \mathbb{E}[\text{flips to reach HHH} \mid \text{current run length} = 0],$$

and similarly define E_1 (last flip was H) and E_2 (last two flips were HH). When we reach HHH, the remaining time is $E_3 = 0$.

From state 0: one flip passes, and with prob. $1/2$ we go to 1 (H), with prob. $1/2$ we stay at 0 (T):

$$E_0 = 1 + \frac{1}{2}E_1 + \frac{1}{2}E_0 \implies E_0 = 2 + E_1.$$

From state 1:

$$E_1 = 1 + \frac{1}{2}E_2 + \frac{1}{2}E_0.$$

From state 2:

$$E_2 = 1 + \frac{1}{2}E_3 + \frac{1}{2}E_0 = 1 + \frac{1}{2}E_0.$$

Substitute E_2 into E_1 :

$$E_1 = 1 + \frac{1}{2}(1 + \frac{1}{2}E_0) + \frac{1}{2}E_0 = 1 + \frac{1}{2} + \frac{1}{4}E_0 + \frac{1}{2}E_0 = \frac{3}{2} + \frac{3}{4}E_0.$$

Then $E_0 = 2 + E_1 = 2 + \frac{3}{2} + \frac{3}{4}E_0 = \frac{7}{2} + \frac{3}{4}E_0$, so

$$\frac{1}{4}E_0 = \frac{7}{2} \implies E_0 = 14.$$

So the expected number of flips to see HHH is $\boxed{14}$.

5. Waiting time for a pattern (THH).

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.1, “Coin triplets, Part A”, book pp. 109–110.

Now the target pattern starts with T, so the natural states are the *longest suffix* of the observed sequence that is also a prefix of THH.

Let:

- E_0 : expected remaining flips from the start state (no useful suffix),
- E_T : expected remaining flips given the last flip is T,
- E_{TH} : expected remaining flips given the last two flips are TH,
- absorbing after THH.

Transitions:

- From start: H gives no prefix (back to start), T gives state T:

$$E_0 = 1 + \frac{1}{2}E_0 + \frac{1}{2}E_T \Rightarrow E_0 = 2 + E_T.$$

- From state T: T keeps you in T; H moves you to TH:

$$E_T = 1 + \frac{1}{2}E_T + \frac{1}{2}E_{TH} \Rightarrow E_T = 2 + E_{TH}.$$

- From state TH: H completes THH (done); T makes the last flip T, so back to state T:

$$E_{TH} = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2}E_T = 1 + \frac{1}{2}E_T.$$

Solve:

$$E_T = 2 + E_{TH} = 2 + 1 + \frac{1}{2}E_T \Rightarrow \frac{1}{2}E_T = 3 \Rightarrow E_T = 6.$$

Then $E_0 = 2 + E_T = 8$. Hence the expected number of flips to see THH is $\boxed{8}$.

6. Expected time to obtain PPF.

Let state 0 mean that no useful suffix is present, state 1 that the current suffix is P, and state 2 that it is PP. Write E_i for the expected number of additional tosses from state i . Completion of PPF has value 0.

Conditioning on the next toss gives

$$\begin{aligned} E_0 &= 1 + \frac{1}{2}E_1 + \frac{1}{2}E_0, \\ E_1 &= 1 + \frac{1}{2}E_2 + \frac{1}{2}E_0, \\ E_2 &= 1 + \frac{1}{2}E_2 + \frac{1}{2} \cdot 0. \end{aligned}$$

The last equation reflects the overlap: after PPP, the last two letters are still PP. Equivalently,

$$E_0 = 2 + E_1, \quad E_1 = 1 + \frac{1}{2}E_2 + \frac{1}{2}E_0, \quad E_2 = 2.$$

Thus $E_1 = 2 + \frac{1}{2}E_0$ and

$$E_0 = 4 + \frac{1}{2}E_0, \quad \text{so} \quad \boxed{E_0 = 8}.$$

7. Expected time to obtain 564 with a die.

Use states 0, 5, and 56, according to the longest suffix that is a prefix of the target word 564. Let the corresponding expected remaining times be E_0, E_5, E_{56} . First-step analysis yields

$$\begin{aligned} E_0 &= 1 + \frac{1}{6}E_5 + \frac{5}{6}E_0, \\ E_5 &= 1 + \frac{1}{6}E_{56} + \frac{1}{6}E_5 + \frac{4}{6}E_0, \\ E_{56} &= 1 + \frac{1}{6} \cdot 0 + \frac{1}{6}E_5 + \frac{4}{6}E_0. \end{aligned}$$

In the second equation, another 5 preserves state 5; in the third, a 5 returns to state 5, while 1, 2, 3, or 6 returns to state 0.

The first equation gives $E_0 = 6 + E_5$. Substitution in the other two equations gives

$$5E_5 = 6 + E_{56} + 4E_0, \quad 6E_{56} = 6 + E_5 + 4E_0.$$

Solving these three linear equations gives

$$E_5 = 210, \quad E_{56} = 180, \quad \boxed{E_0 = 216}.$$

8. DP warm-up: last stage of a 3-roll dice game.

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.3, “Dice game”, book p. 123.

At the third (final) roll you have no choice: you must take the value of a fair die. So the value function at the last stage is simply the mean of a fair die:

$$V_{\text{final}} = \mathbb{E}[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5.$$

9. DP threshold after the second roll (3-roll dice game).

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.3, “Dice game”, book p. 123.

After the second roll you observe $x \in \{1, \dots, 6\}$ and choose:

$$\text{Stop: get } x \quad \text{vs} \quad \text{Continue: expected value } V_{\text{final}} = 3.5.$$

Therefore:

$$\text{Stop iff } x \geq 3.5 \quad \Longleftrightarrow \quad x \in \{4, 5, 6\}.$$

For $x \in \{1, 2, 3\}$ you should continue.

Medium (9)

10. DP threshold after the first roll (3-roll dice game).

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.3, “Dice game”, book p. 123.

Work backwards.

Step 1: value at the second-roll stage before seeing the second roll. Let Y be the second roll. Under the optimal rule from the previous exercise,

if $Y \in \{4, 5, 6\}$ stop and take Y , if $Y \in \{1, 2, 3\}$ continue and get 3.5.

Thus the *continuation value* at the start of roll 2 is

$$V_2 = \frac{4+5+6}{6} + \frac{3}{6} \cdot 3.5 = \frac{15}{6} + \frac{3.5}{2} = 2.5 + 1.75 = 4.25 = \frac{17}{4}.$$

Step 2: decision after the first roll. After the first roll you see x and choose stop (get x) or continue (expected $V_2 = 4.25$). So:

$$\text{Stop iff } x \geq 4.25 \iff x \in \{5, 6\}.$$

11. Dice race: 12 vs two consecutive 7s.

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.1, “Dice question”, book pp. 108–109.

Let $a = \mathbb{P}(12) = \frac{1}{36}$, $b = \mathbb{P}(7) = \frac{6}{36} = \frac{1}{6}$, and $c = \mathbb{P}(\text{other}) = \frac{29}{36}$.

Define two states:

- State 0: the previous roll was *not* 7 (so B has no current streak),
- State 1: the previous roll *was* 7 (so B has a streak of length 1).

Let p_0 be the probability A eventually wins starting from state 0, and p_1 similarly from state 1.

From state 0:

$$p_0 = a \cdot 1 + b \cdot p_1 + c \cdot p_0 \implies (1 - c)p_0 = a + bp_1.$$

Since $1 - c = 7/36$, this becomes

$$\frac{7}{36}p_0 = \frac{1}{36} + \frac{1}{6}p_1 \implies 7p_0 = 1 + 6p_1.$$

From state 1:

$$p_1 = a \cdot 1 + b \cdot 0 + c \cdot p_0 = \frac{1}{36} + \frac{29}{36}p_0 \implies 36p_1 = 1 + 29p_0.$$

Solve the system:

$$p_1 = \frac{1 + 29p_0}{36} \implies 7p_0 = 1 + 6 \cdot \frac{1 + 29p_0}{36} = 1 + \frac{1 + 29p_0}{6}.$$

Multiply by 6:

$$42p_0 = 6 + 1 + 29p_0 \implies 13p_0 = 7 \implies p_0 = \frac{7}{13}.$$

So $\boxed{\mathbb{P}(\text{A wins}) = 7/13}$.

12. Competing patterns: HHH vs THH.

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.1, “Coin triplets, Part B”, book pp. 110–111.

A key observation: *if a T ever occurs before HHH is completed, then THH will occur before HHH.* Why? Because to obtain HHH after some T appears, the first time you see two consecutive H's after that T, you have already formed THH.

So the only way to see HHH *before* THH is to start with HHH immediately:

$$\mathbb{P}(\text{HHH occurs before THH}) = \mathbb{P}(\text{first three flips are HHH}) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}.$$

Thus $\boxed{1/8}$.

13. Ticket line probability.

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.2, “Ticket line”, book pp. 117–118.

Encode the queue as a word of length $2n$ consisting of n symbols “5” and n symbols “10”. The cashier can always sell tickets if and only if, in every prefix of the word,

$$\#(5\text{'s so far}) \geq \#(10\text{'s so far}),$$

because each 10-dollar customer requires one 5-dollar bill previously collected as change.

The number of such valid sequences is the *Catalan number*

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

Total sequences (all orders) is $\binom{2n}{n}$. Therefore the probability is

$$\mathbb{P}(\text{everyone can buy}) = \frac{C_n}{\binom{2n}{n}} = \frac{1}{n+1}.$$

So the answer is $\boxed{\frac{1}{n+1}}$.

14. Expected time to get n heads in a row.

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.2, “Coin sequence”, book p. 119.

Let T_n be the number of flips needed to see n consecutive heads. Define states by the current run length of consecutive heads: $k \in \{0, 1, \dots, n\}$. Let $E_k = \mathbb{E}[T_n \mid \text{current run length} = k]$. We want E_0 .

Boundary: $E_n = 0$. For $k < n$,

$$E_k = 1 + \frac{1}{2}E_{k+1} + \frac{1}{2}E_0,$$

because on the next flip: with probability $1/2$ we add one head (run length increases to $k+1$), and with probability $1/2$ we get a tail, which resets the run length to 0.

Rearrange:

$$E_k - \frac{1}{2}E_{k+1} = 1 + \frac{1}{2}E_0.$$

Solve by backward substitution. First, with $k = n - 1$:

$$E_{n-1} = 1 + \frac{1}{2}E_n + \frac{1}{2}E_0 = 1 + \frac{1}{2}E_0.$$

With $k = n - 2$:

$$E_{n-2} = 1 + \frac{1}{2}E_{n-1} + \frac{1}{2}E_0 = 1 + \frac{1}{2}\left(1 + \frac{1}{2}E_0\right) + \frac{1}{2}E_0 = \frac{3}{2} + \frac{3}{4}E_0.$$

Continuing this pattern yields

$$E_{n-m} = (2 - 2^{-m+1}) + (1 - 2^{-m+1})E_0,$$

and plugging $m = n$ (i.e. E_0 on the left) gives the closed form

$$E_0 = 2^{n+1} - 2.$$

Hence

$$\boxed{\mathbb{E}[T_n] = 2^{n+1} - 2.}$$

(Check: for $n = 1$, gives 2; for $n = 3$, gives 14, matching the earlier result.)

15. A pattern with overlap (PFP).

Let E_0, E_P, E_{PF} denote the expected remaining tosses when the longest useful suffix is respectively empty, P, or PF. First-step conditioning gives

$$\begin{aligned} E_0 &= 1 + \frac{1}{2}E_P + \frac{1}{2}E_0, \\ E_P &= 1 + \frac{1}{2}E_P + \frac{1}{2}E_{PF}, \\ E_{PF} &= 1 + \frac{1}{2} \cdot 0 + \frac{1}{2}E_0. \end{aligned}$$

The transition $P \xrightarrow{P} P$ keeps one matched letter. By contrast, $PF \xrightarrow{F} 0$, since PFF has no nonempty suffix that is a prefix of PFP.

The system is equivalent to

$$E_0 = 2 + E_P, \quad E_P = 2 + E_{PF}, \quad E_{PF} = 1 + \frac{1}{2}E_0.$$

Consequently,

$$E_0 = 5 + \frac{1}{2}E_0 \quad \implies \quad \boxed{E_0 = 10}.$$

The answer exceeds $2^3 = 8$ because PFP overlaps with itself through its final P; a mismatch can preserve partial progress, but it also creates longer clusters of possible completion times.

16. World Series (fair games): replication via DP.

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.3, “World series”, book pp. 123–125.

Let $f(i, j)$ be the *replication value* (wealth you should hold right now) when Team A has i wins and Team B has j wins. The series ends when someone reaches 4 wins.

Terminal payoffs:

$$f(4, j) = 100, \quad f(i, 4) = -100.$$

For a fair game, the next game is a 50/50 coin flip. Therefore the DP pricing recursion is

$$f(i, j) = \frac{f(i+1, j) + f(i, j+1)}{2}.$$

Replicating bet. Suppose you bet $y(i, j)$ dollars on Team A in the next game with even odds. If A wins, wealth increases by $+y$; if A loses, wealth decreases by y . To replicate the continuation values we need

$$f(i, j) + y(i, j) = f(i+1, j), \quad f(i, j) - y(i, j) = f(i, j+1).$$

Solving gives

$$y(i, j) = \frac{f(i+1, j) - f(i, j+1)}{2}.$$

Compute $y(2, 1)$. We compute $f(3, 1)$ and $f(2, 2)$ first (working backward from the boundary). One finds:

$$f(3, 1) = 75, \quad f(2, 2) = 0.$$

Hence

$$y(2, 1) = \frac{75 - 0}{2} = 37.5.$$

Interpretation: when the score is $(2, 1)$, bet \$37.50 on Team A next game.

17. Dynamic dice game (optimal stopping with wipeout).

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.3, “Dynamic dice game”, book p. 126.

Let n be your current accumulated winnings. If you stop, you lock in n . If you continue for one more roll, then:

- with probability $1/6$ you roll 6 and end with 0,
- with probability $1/6$ you roll $k \in \{1, 2, 3, 4, 5\}$ and move to state $n + k$.

Let $V(n)$ be the optimal expected payoff from state n . Then the Bellman equation is

$$V(n) = \max\left(n, \underbrace{\frac{1}{6} \sum_{k=1}^5 V(n+k)}_{\text{continue value}}\right),$$

(where the roll-6 outcome contributes 0 and is already included by the $1/6$ weight).

Threshold intuition. If you continue one more step and *then* stop immediately, your expected wealth after that roll would be

$$\frac{1}{6}(n+1) + \cdots + \frac{1}{6}(n+5) + \frac{1}{6} \cdot 0 = \frac{5}{6}n + 2.5.$$

Continuing is attractive when this exceeds n :

$$\frac{5}{6}n + 2.5 > n \iff n < 15.$$

So it is optimal to continue for $n \leq 14$ and (at least) stop for $n \geq 15$.

Compute the entry price $V(0)$. Using the threshold and backward recursion (as in the book), one obtains $V(0) \approx 6.15$. So a risk-neutral player is willing to pay about

$$\boxed{\$6.15}$$

to enter the game, and the optimal policy is “**keep rolling until you have \$15 or more, then stop.**”

18. Optimal stopping (3-step) — alternative statement.

Reference: Heard on the Street, Ch. 4 “Statistics Questions”, Question 4.4, book p. 37.

This is the same *finite-horizon optimal stopping* as the 3-roll dice game above.

Step 3 (last roll): value is 3.5.

Step 2: stop if $x \geq 3.5$, i.e. stop for $x \in \{4, 5, 6\}$.

Step 1: if you continue from roll 1, your expected value is the roll-2 value $V_2 = 4.25$ computed earlier. So you stop after the first roll iff $x \geq 4.25$, i.e. $x \in \{5, 6\}$.

Game value. At roll 1, if $x \in \{1, 2, 3, 4\}$ you continue (value 4.25), if $x \in \{5, 6\}$ you stop and take x . Therefore

$$V_1 = \frac{4}{6} \cdot 4.25 + \frac{5+6}{6} = \frac{17}{6} + \frac{11}{6} = \frac{28}{6} = \frac{14}{3} \approx 4.667.$$

So the optimal expected payoff is $\boxed{14/3}$.

Hard (5)

19. Penney’s game with strategic choice (who wants to go first?).

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.1, “Coin triplets, Part C”, book pp. 111–112.

Conclusion: you prefer to be Player 2. Penney’s game is *non-transitive*: there is no single “best” pattern.

Player 2’s winning strategy (length 3). Suppose Player 1 chooses a pattern ABC (each letter is H or T). A standard best response for Player 2 is

$$\boxed{\overline{B} A B}$$

where $\overline{B}$ is the *opposite* of B (i.e. $\overline{H} = T$, $\overline{T} = H$).

Why this works (intuition). Player 2’s pattern is built so that whenever the coin sequence begins to look like Player 1’s pattern, it is also “primed” to complete Player 2’s pattern sooner due to overlaps of length 2.

A concrete example is Player 1 choosing HHH. Then Player 2 chooses THH, and indeed

$$\mathbb{P}(\text{THH appears before HHH}) = 1 - \mathbb{P}(\text{HHH before THH}) = 1 - \frac{1}{8} = \frac{7}{8}.$$

Guaranteed probability. Across all Player 1 choices, Player 2 can always pick a response with winning probability strictly greater than $1/2$. Moreover, the *worst case* for Player 2 (with optimal play) is

$\text{Player 2 can guarantee at least } \frac{2}{3} \text{ to win.}$

So you would rather be Player 2.

20. Color balls (expected absorption time).

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.1, “Color balls”, book pp. 113–115.

Let N_n be the number of steps until all n balls are the same color, starting from n distinct colors.

Step 1: symmetry to fix the final color. Let F_i be the event that the final consensus color is color i . By symmetry, $\mathbb{P}(F_i) = 1/n$ and $\mathbb{E}[N_n | F_i]$ is the same for all i . Hence

$$\mathbb{E}[N_n] = \mathbb{E}[N_n | F_1].$$

Step 2: reduce to a 1D Markov chain. Condition on F_1 (the process ends with all balls color 1). Let X_k be the number of balls of color 1 after k steps (under this conditioning). Then $X_k \in \{1, 2, \dots, n\}$ and n is absorbing.

Under the conditioned chain, one can compute the transition probabilities (see the book) and set up the standard mean-time-to-absorption recursion for

$$Z_i := \mathbb{E}[N_n | F_1, X_0 = i].$$

With boundary $Z_n = 0$ and a second-order recursion, the system solves to

$$Z_1 = (n-1)^2.$$

Since $X_0 = 1$ in the initial “all different” configuration (exactly one ball is color 1),

$\mathbb{E}[N_n] = Z_1 = (n-1)^2.$

21. Random walk as a Markov chain (bridge problem).

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.2, “Drunk man”, book pp. 116–117.

Let X_t be the drunk’s position on $\{0, 1, \dots, 100\}$ with absorbing boundaries at 0 and 100. He starts at $X_0 = 17$ and moves ± 1 with equal probability.

(i) Probability to hit 100 before 0. For a symmetric random walk with absorbing endpoints, the hitting probability is linear:

$$h(i) = \mathbb{P}_i(\text{hit 100 before 0})$$

satisfies $h(0) = 0$, $h(100) = 1$, and for $1 \leq i \leq 99$,

$$h(i) = \frac{1}{2}h(i-1) + \frac{1}{2}h(i+1).$$

The unique solution is $h(i) = i/100$, hence

$$\mathbb{P}(\text{reach 100 first}) = h(17) = \frac{17}{100}.$$

(ii) Expected absorption time. Let $m(i) = \mathbb{E}_i[\tau]$ where τ is the hitting time of $\{0, 100\}$. Then $m(0) = m(100) = 0$ and for $1 \leq i \leq 99$,

$$m(i) = 1 + \frac{1}{2}m(i-1) + \frac{1}{2}m(i+1).$$

This second-order difference equation has solution $m(i) = i(100-i)$ for the symmetric case. Therefore

$$\mathbb{E}[\tau \mid X_0 = 17] = 17 \cdot (100 - 17) = 17 \cdot 83 = 1411.$$

22. Dynamic card game (DP / optimal stopping in finite deck).

Reference: A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.3, “Dynamic card game”, book pp. 127–128.

There are 26 red cards (+\$1 each) and 26 black cards (-\$1 each), drawn without replacement. Let (b, r) be the number of black and red cards *remaining* in the deck.

A convenient state variable for current payoff is

$$(\text{red drawn}) - (\text{black drawn}) = (26 - r) - (26 - b) = b - r.$$

So, if you stop when the remaining deck is (b, r) , your payoff is $b - r$.

Let $V(b, r)$ be the optimal expected payoff from state (b, r) . At (b, r) you either stop (get $b - r$) or continue. If you continue:

- with probability $\frac{b}{b+r}$ you draw a black card, gain -1 and move to $(b-1, r)$,
- with probability $\frac{r}{b+r}$ you draw a red card, gain $+1$ and move to $(b, r-1)$.

Because the payoff $b - r$ already tracks the accumulated score, the Bellman recursion can be written directly as in the book:

$$V(b, r) = \max\left(b - r, \frac{b}{b+r} V(b-1, r) + \frac{r}{b+r} V(b, r-1)\right).$$

Boundary conditions:

$$V(0, r) = 0,$$

because, when only red cards remain, continuing to the end yields 0 in the $b - r$ convention. Also,

$$V(b, 0) = b.$$

Running the recursion (or using the figure/table in the book) gives

$$V(26, 26) \approx 2.62.$$

So a risk-neutral player should be willing to pay about \$2.62 to play, under optimal stopping.

23. **World Series with biased win probabilities (DP hedging).**

Reference: Based on A Practical Guide to Quantitative Finance Interviews, Ch. 5 §5.3, “World series” (fair case shown), book pp. 123–125.

Let $f(i, j)$ be the replication value at series state (i, j) (A has i wins, B has j). Terminal values stay the same:

$$f(4, j) = 100, \quad f(i, 4) = -100.$$

Assume each game is independent and A wins with probability p . If the bet you can place each game is *fairly priced* (zero expected gain), then the no-arbitrage DP pricing recursion becomes

$$\boxed{f(i, j) = p f(i + 1, j) + (1 - p) f(i, j + 1).}$$

To replicate the claim, choose a stake $y(i, j)$ on Team A in the next game so that the post-game wealth matches the continuation values. With fair odds, “winning” a bet of size y should pay $\frac{1-p}{p} y$ (so that expected profit is zero), and “losing” costs y . Thus we impose

$$f(i, j) + \frac{1-p}{p} y(i, j) = f(i + 1, j), \quad f(i, j) - y(i, j) = f(i, j + 1).$$

Solving:

$$\boxed{y(i, j) = p(f(i + 1, j) - f(i, j + 1)), \quad f(i, j) = p f(i + 1, j) + (1 - p) f(i, j + 1).}$$

(These are the standard “value” and “hedge” equations on a binomial tree.)